

WT EXPERIMENT - 05

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Aim:

To simulate a project using Wokwi, an online electronics simulator, focusing on an RFID-based access control system with a servo barrier and LCD display.

Theory:

Wokwi is a cloud-based platform designed for simulating electronics projects, particularly those involving microcontrollers like Arduino, ESP32, and STM32. It offers a user-friendly interface for designing and testing circuits without the need for physical hardware, making it ideal for beginners and experienced makers alike.

Overview of Wokwi

- Type of Site: Electronics simulation and design
- Owner: Wokwi
- URL: www.wokwi.com
- Commercial: Offers free and paid plans
- Registration: Required for full features
- Key Features:
 - Simulation: Supports real-time simulation of Arduino and other microcontroller projects.
 - Components: Includes a wide range of components such as LEDs, sensors, and actuators.
 - Collaboration: Allows users to share and collaborate on projects in real-time.
 - Debugging: Offers advanced debugging tools like GDB for Arduino and Raspberry Pi Pico.

Code:

The following code is for an RFID-based access control system using an Arduino board, a servo barrier, an LCD display, and a buzzer.

```
cpp
```

```
#include <Wire.h>
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```
#include <LiquidCrystal_I2C.h>
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#include <Servo.h>

#define BUZZER_PIN 2
#define SERVO_PIN 7
#define VALID_TAG "12345" // Simulated RFID Tag

LiquidCrystal_I2C lcd(0x27, 16, 2); // LCD Address (Change if needed)
Servo barrierServo;

void setup() {
    Serial.begin(9600);
    lcd.begin(16, 2);
    lcd.backlight();

    pinMode(BUZZER_PIN, OUTPUT);
    barrierServo.attach(SERVO_PIN);
    barrierServo.write(0); // Start with barrier closed

    lcd.setCursor(0, 0);
    lcd.print("Scan your card");
}

void loop() {
    if (Serial.available()) {
        String scannedTag = Serial.readStringUntil('\n');
        scannedTag.trim(); // Remove spaces/newlines

        lcd.clear();
        if (scannedTag == VALID_TAG) {
            lcd.setCursor(0, 0);
            lcd.print("Access Granted");

            digitalWrite(BUZZER_PIN, HIGH);
            delay(500);
            digitalWrite(BUZZER_PIN, LOW);

            barrierServo.write(90); // Open barrier
            delay(3000);
            barrierServo.write(0); // Close barrier
        } else {
            lcd.setCursor(0, 0);
            lcd.print("Access Denied");

            digitalWrite(BUZZER_PIN, HIGH);
            delay(1000);
            digitalWrite(BUZZER_PIN, LOW);
        }
    }
}

```

```

    }

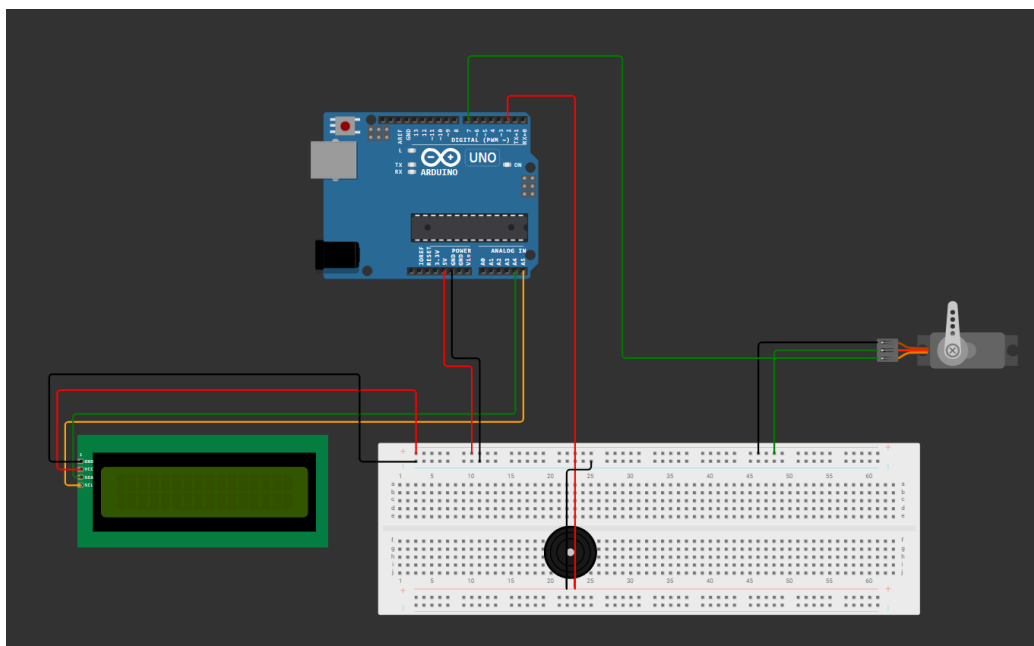
    delay(2000);
    lcd.clear();
    lcd.print("Scan your card");
  }
}

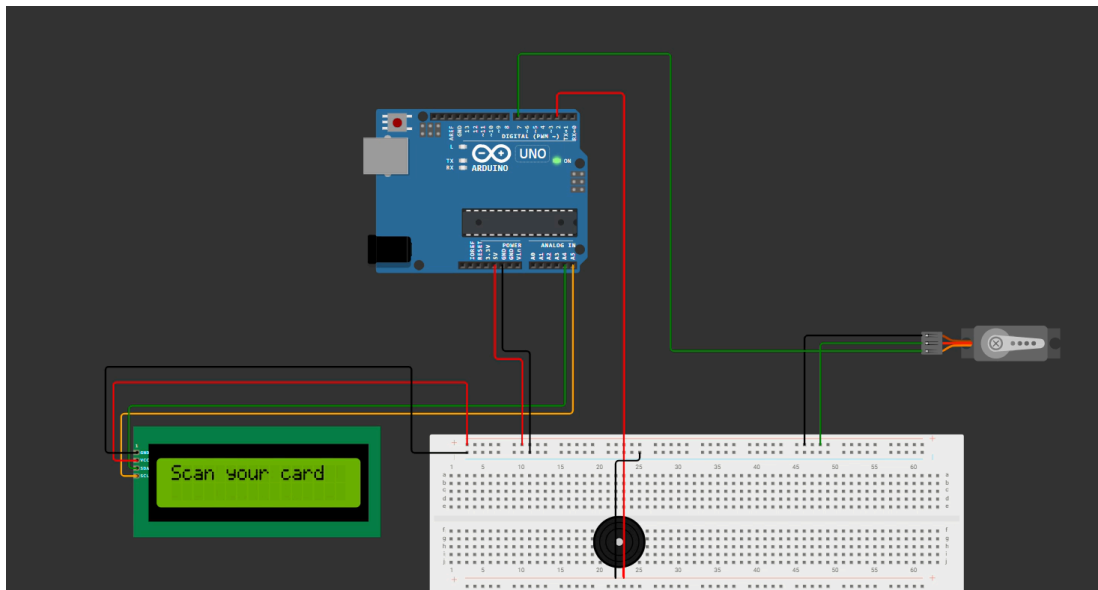
```

Practical Steps:

1. Setting Up the Project in Wokwi:
 - Go to www.wokwi.com and create a new project.
 - Choose the Arduino board you wish to simulate (e.g., Arduino Uno).
 - Add components to your project, including an LCD display, servo motor, buzzer, and a simulated RFID reader.
2. Writing and Uploading the Code:
 - Write the code as shown above.
 - Upload the code to your virtual Arduino board in Wokwi.
 - Simulate the RFID tag scan by sending the VALID_TAG string through the serial monitor.
3. Testing the Simulation:
 - Observe the LCD display for "Access Granted" or "Access Denied" messages.
 - Verify that the servo barrier opens and closes accordingly.
 - Check the buzzer for the appropriate sound signals.

Simulation:





Conclusion:

This project demonstrates how to use Wokwi for simulating an RFID-based access control system with a servo barrier and LCD display. The simulation allows for testing and debugging without physical hardware, making it an efficient tool for prototyping and educational purposes.